Nicklas Hansen

Postdoctoral Associate, NYU Courant
World models, reinforcement learning, and robot learning
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Experience

New York University, Courant Institute — New York, NY
Postdoctoral Associate — *Aug 2026 – present*
Supervisors: Lerrel Pinto and Yann LeCun.

NVIDIA Research — Seattle, WA
Research Intern — *Jun 2024 – Feb 2025*
Supervisors: Yashraj Narang and Dieter Fox.

Meta AI (FAIR) — Menlo Park, CA
Research Scientist Intern — *Jun 2022 – Dec 2022*
Supervisor: Aravind Rajeswaran.

University of California, San Diego — San Diego, CA
Research Intern — *Jul 2020 – Aug 2021*
Supervisor: Xiaolong Wang.

Education

University of California, San Diego — San Diego, CA
PhD, Computer Science and Engineering — *Sep 2021 – Jun 2026*
Advised by Xiaolong Wang and Hao Su. NVIDIA Graduate Fellow.

University of California, Berkeley — Berkeley, CA
Visiting Student — *Spring 2020*
Spar Nord Fonden FinTech Scholarship recipient. SCET's Collider Cup finalist.

Technical University of Denmark — Kongens Lyngby, Denmark
MS, Mathematical Modelling and Computation — *Feb 2019 – Feb 2021*
Advised by Ole Winther.

Technical University of Denmark — Kongens Lyngby, Denmark
BS, Software Technology — *Sep 2015 – Dec 2018*
Semester abroad at Nanyang Technological University, Singapore, Fall 2017.

Fellowships and Grants

2026 **Nebius Research Grant** — *Sole PI, $35,500*
Cloud compute credits for a controlled comparison of world models.

2024 **NVIDIA Graduate Fellowship** — *$60,000*
Stipend and tuition for one year (10 recipients worldwide).

Honors and Awards

2026 **UC Riverside Future Academics & Mentorship Excellence (FAME) Program**
A three-day workshop for future faculty.

2024 **MyoSuite Symposium at NeurIPS**
2nd place in the Locomotion track and the Physiological Award, for musculoskeletal control with TD-MPC2.

2023 **NeurIPS Top Reviewer**

2017–21 **Scholarships**
Robotics Summer School Scholarship (2021); Spar Nord Fonden FinTech Scholarship for a semester at UC Berkeley, 5 recipients nationwide (2020); Otto Mønsted Fonds Legat for a semester abroad (2017).

Peer-Reviewed Publications

★ Representative papers. Underlined: students I mentored. * and † mark equal contribution or equal advising.

Conference Papers

[C29] **Hallucination in World Models is Predictable and Preventable** ★
Nicklas Hansen, Xiaolong Wang
Conference on Neural Information Processing Systems (NeurIPS), 2026 link

[C28] **Abstracting Task-Centric World Models from Visual Foundations**
Minghao Fu, Fan Feng, **Nicklas Hansen**, Biwei Huang
Conference on Neural Information Processing Systems (NeurIPS), 2026 link

[C27] **FACT: Failure-Aware Causal Training for World-Action Models**
Quanquan Peng*, Yutong Liang*, Rui Yan, **Nicklas Hansen**, Xiaolong Wang
Conference on Robot Learning (CoRL), 2026 link

[C26] **Generalizable Robotic Insertion with World Models**
Nicklas Hansen, Iretiayo Akinola, Yijie Guo, Jie Xu, Bingjie Tang, Hao Su, Xiaolong Wang, Abhishek Gupta, Dieter Fox, Yashraj Narang
International Conference on Intelligent Robots and Systems (IROS), 2026 link

[C25] **Learning Reasoning-Guided Robot Policies via Task-Aligned Simulated Futures**
Hanan Gani, Tejal Kulkarni, Madhoolika Chodavarapu, **Nicklas Hansen**, Manmohan Chandraker
European Conference on Computer Vision (ECCV), 2026 link

[C24] **Learning to Design Soft Hands using Reward Models**
Xueqian Bai, **Nicklas Hansen**, Adabhav Singh, Michael T. Tolley, Yan Duan, Pieter Abbeel, Xiaolong Wang, Sha Yi
International Conference on Robotics and Automation (ICRA), 2026 link

[C23] **Learning Massively Multitask World Models for Continuous Control** ★
Nicklas Hansen, Hao Su†, Xiaolong Wang†
International Conference on Learning Representations (ICLR), 2026 link

[C22] **TIPS: Turn-Level Information-Potential Reward Shaping for Search-Augmented LLMs**
Yutao Xie, Nathaniel Thomas, **Nicklas Hansen**, Yang Fu, Li Erran Li, Xiaolong Wang
International Conference on Learning Representations (ICLR), 2026 link

[C21] **MyoChallenge 2024: A New Benchmark for Physiological Dexterity and Agility in Bionic Humans**
Cheryl Wang, Chun Kwang Tan, [...], **Nicklas Hansen**, [...], Vikash Kumar, Vittorio Caggiano (26 authors)
Conference on Neural Information Processing Systems (NeurIPS), 2025 link

[C20] **Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning**
Adrià López Escoriza, **Nicklas Hansen**, Stone Tao, Tongzhou Mu, Hao Su
International Conference on Machine Learning (ICML), 2025 link

[C19] **Hierarchical World Models as Visual Whole-Body Humanoid Controllers**
Nicklas Hansen, Jyothir S V, Vlad Sobal, Yann LeCun, Xiaolong Wang*, Hao Su*
International Conference on Learning Representations (ICLR), 2025 link

[C18] **Policy Learning with Multi-Task World Models**
Ignat Georgiev, Varun Giridhar, **Nicklas Hansen**, Animesh Garg
International Conference on Learning Representations (ICLR), 2025 link

[C17] **A Simulation Benchmark for Autonomous Racing with Large-Scale Human Data**
Adrian Remonda, **Nicklas Hansen**, Ayoub Raji, Nicola Musiu, Marko Bertogna, Eduardo E. Veas, Xiaolong Wang
Conference on Neural Information Processing Systems (NeurIPS), 2024 link

[C16] **A Recipe for Unbounded Data Augmentation in Visual Reinforcement Learning**
Abdulaziz Almuzairee, **Nicklas Hansen**, Henrik I. Christensen
Reinforcement Learning Conference (RLC), 2024 link

[C15] **TD-MPC2: Scalable, Robust World Models for Continuous Control** ★
Nicklas Hansen, Hao Su†, Xiaolong Wang†
International Conference on Learning Representations (ICLR), 2024 link *Spotlight (top 6.2%)*

[C14] **MoDem-V2: Visuo-Motor World Models for Real-World Robot Manipulation**
Patrick Lancaster, **Nicklas Hansen**, Aravind Rajeswaran, Vikash Kumar
International Conference on Robotics and Automation (ICRA), 2024 link

[C13] **Open X-Embodiment: Robotic Learning Datasets and RT-X Models**
Open X-Embodiment Collaboration, [...], **Nicklas Hansen**, [...]
International Conference on Robotics and Automation (ICRA), 2024 link *Best Conference Paper*

[C12] **Finetuning Offline World Models in the Real World** ★
Yunhai Feng*, **Nicklas Hansen***, Ziyan Xiong*, Chandramouli Rajagopalan, Xiaolong Wang
Conference on Robot Learning (CoRL), 2023 link *Oral (top 6.6%)*

[C11] **Multi-Task Real Robot Learning with Generalizable Neural Feature Fields**
Yanjie Ze, Ge Yan, Yueh-Hua Wu, Annabella Macaluso, Yuying Ge, Jianglong Ye, **Nicklas Hansen**, Li Erran Li, Xiaolong Wang
Conference on Robot Learning (CoRL), 2023 link *Oral (top 6.6%)*

[C10] **On Pre-Training for Visuo-Motor Control: Revisiting a Learning-from-Scratch Baseline**
Nicklas Hansen*, Zhecheng Yuan*, Yanjie Ze*, Tongzhou Mu*, Aravind Rajeswaran†, Hao Su†, Huazhe Xu†, Xiaolong Wang†
International Conference on Machine Learning (ICML), 2023 link

[C9] **MoDem: Accelerating Visual Model-Based Reinforcement Learning with Demonstrations**
Nicklas Hansen, Yixin Lin, Hao Su, Xiaolong Wang, Vikash Kumar, Aravind Rajeswaran
International Conference on Learning Representations (ICLR), 2023 link

[C8] **On the Feasibility of Cross-Task Transfer with Model-Based Reinforcement Learning**
Yifan Xu*, **Nicklas Hansen***, Zirui Wang, Yung-Chieh Chan, Hao Su, Zhuowen Tu
International Conference on Learning Representations (ICLR), 2023 link

[C7] **Graph Inverse Reinforcement Learning from Diverse Videos**
Sateesh Kumar, Jonathan Zamora*, **Nicklas Hansen***, Rishabh Jangir, Xiaolong Wang
Conference on Robot Learning (CoRL), 2022 link *Oral (top 7%)*

[C6] **Temporal Difference Learning for Model Predictive Control** ★
Nicklas Hansen, Xiaolong Wang*, Hao Su*
International Conference on Machine Learning (ICML), 2022 link *Spotlight*

[C5] **Learning Vision-Guided Quadrupedal Locomotion End-to-End with Cross-Modal Transformers**
Ruihan Yang*, Minghao Zhang*, **Nicklas Hansen**, Huazhe Xu, Xiaolong Wang
International Conference on Learning Representations (ICLR), 2022 link *Spotlight (top 6.8%)*

[C4] **Stabilizing Deep Q-Learning with ConvNets and Vision Transformers under Data Augmentation**
Nicklas Hansen, Hao Su, Xiaolong Wang
Conference on Neural Information Processing Systems (NeurIPS), 2021 link

[C3] **Generalization in Reinforcement Learning by Soft Data Augmentation**
Nicklas Hansen, Xiaolong Wang
International Conference on Robotics and Automation (ICRA), 2021 link

[C2] **Self-Supervised Policy Adaptation during Deployment**
Nicklas Hansen, Rishabh Jangir, Yu Sun, Guillem Alenyà, Pieter Abbeel, Alexei A. Efros, Lerrel Pinto, Xiaolong Wang
International Conference on Learning Representations (ICLR), 2021 link *Spotlight (top 5.5%)*

[C1] **Short Term Blood Glucose Prediction Based on Continuous Glucose Monitoring Data**
Ali Mohebbi, Alexander R. Johansen, **Nicklas Hansen**, Peter E. Christensen, Jens M. Tarp, Morten L. Jensen, Henrik Bengtsson, Morten Mørup
IEEE Engineering in Medicine and Biology Society Conference (EMBC), 2020 link

Journal Papers

[J2] **Visual Reinforcement Learning with Self-Supervised 3D Representations**
Yanjie Ze*, **Nicklas Hansen***, Yinbo Chen, Mohit Jain, Xiaolong Wang
IEEE Robotics and Automation Letters (RA-L), 2023 link

[J1] **Look Closer: Bridging Egocentric and Third-Person Views with Transformers for Robotic Manipulation**
Rishabh Jangir*, **Nicklas Hansen***, Sambaran Ghosal, Mohit Jain, Xiaolong Wang
IEEE Robotics and Automation Letters (RA-L), 2022 link

Under Review and Preprints

[P7] **Bayesian Curriculum Learning over Latent Geometry in Large Language Models**
Darrien McKenzie, **Nicklas Hansen**†, Xiaolong Wang†
Under review, 2026 link

[P6] **Catch It All!: Generalizable Dynamic Catching with a Dexterous Hand**
Arth Shukla*, Kehlani Fay*, Huaze Liu†, Darin Djapri†, Alexiy Buynitsky, **Nicklas Hansen**
Under review, 2026

[P5] **Generating Robot Hands from Human Demonstrations**
Sha Yi, **Nicklas Hansen**, Xueqian Bai, Carmelo Sferrazza, Michael T. Tolley, Xiaolong Wang
Under review, 2026 link

[P4] **HiGym: Learning to Act in High-Dimensional Spaces**
Harshit Dhankhar, Xiaolong Wang, **Nicklas Hansen**
Under review, 2026

[P3] **Simulation Priors for Rapid Adaptation in Model-Based RL for Autonomous Racing**
Adrian Remonda, Jiajun Xi, **Nicklas Hansen**, Marcus Greiff, Thomas Lew, John Talbot, John Subosits, Xiaolong Wang
Under review, 2026

[P2] **SNAP: One-Shot Dexterous Manipulation via Functional Grasping Policies**
Alexiy Buynitsky, **Nicklas Hansen**, Carmelo Sferrazza, Jianglong Ye, Xiaolong Wang
Under review, 2026

[P1] **Trust Before Use: When Do World-Model Quality Metrics Predict Control?**
Varun Giridhar, Hrishit Leen, Ignat Georgiev, **Nicklas Hansen**, Animesh Garg
Under review, 2026

Open-Source Software

2026 **Dreamer 4, unofficial implementation** ☆ 408
PyTorch implementation of the Dreamer 4 world model.

2024 **TD-MPC2, official implementation** ☆ 976
Code release for "TD-MPC2: Scalable, Robust World Models for Continuous Control."

2022 **TD-MPC, official implementation** ☆ 527
Code release for "Temporal Difference Learning for Model Predictive Control."

Mentoring

35 students since 2020 (12 current); 9 now in PhD programs; first or co-first authors on 3 CoRL orals.

Former (23)

Rishabh Jangir (MS, UCSD → ML Engineer, Skild AI)	2020 – 2022
Mohit Jain (MS, UCSD → ML Engineer, Pinterest)	2020 – 2022
Xinyue Chen (BS, NYU Shanghai → PhD, UC Berkeley)	2021 – 2022
Sateesh Kumar (MS, UCSD → PhD, UT Austin)	2021 – 2023
Jonathan Zamora (BS, UCSD → MS, USC)	2021 – 2023
Sambaran Ghosal (MS, UCSD → ML Engineer, BrainCorp)	2021 – 2023
Yanjie Ze (BS, SJTU → PhD, Stanford)	2021 – 2023
Zirui "Colin" Wang (BS, UCSD → PhD, UC Berkeley)	2022 – 2023
Ziyan Xiong (BS, Tsinghua → PhD, UMass)	2022 – 2023
Yunhai Feng (MS, UCSD → PhD, Cornell)	2022 – 2024
Chandramouli Rajagopalan (MS, UCSD → ML Engineer, Streamingo)	2022 – 2024
Jyothir S V (MS, NYU → RA, NYU)	2023 – 2024
Adrià López Escoriza (MS, ETH → PhD, ETH)	2024 – 2025
Varun Giridhar (MS, Georgia Tech → RA, Georgia Tech)	2024 – 2025
Nathaniel Thomas (MS, UCSD → ML Engineer, Meta AI)	2024 – 2026
Yutao Xie (MS, UCSD → PhD, UC Santa Barbara)	2024 – 2026
Jiajun Xi (MS, UCSD)	2024 – 2026
Adrian Remonda (PhD, TU Graz → Research Scientist, TRI)	2024 – 2026
Xueqian Bai (MS, UCSD → Researcher, XPeng)	2025 – 2026
Darrien McKenzie (PhD, UCSD)	2025 – 2026
Arth Shukla (BS, UCSD → PhD, UW)	2026
Kehlani Fay (PhD, UCSD)	2026

Harshit Dhankhar (MS, UCSD) 2026

Current (12)

Jai Bandi (MS, UCSD) 2025 – present
Sachin Bhadang (MS, UCSD) 2026 – present
Alexiy Buynitsky (MS, UCSD) 2026 – present
Madhoolika Chodavarapu (MS, UCSD) 2026 – present
Zavier Andrianarivo (RA, NYU) 2026 – present
Enoch Wang (BS, NYU) 2026 – present
Lakshya Jain (BS, NYU) 2026 – present
Huayuan Hu (BS, NYU) 2026 – present
Charles Ji (BS, NYU) 2026 – present
Henry He (BS, NYU) 2026 – present
Yiyang Zhang (MS, NYU) 2026 – present
Sally Qiao (BS, NYU) 2026 – present

Teaching

University of California, San Diego San Diego, CA
Teaching Assistant, ECE 285: Introduction to Visual Learning *Spring 2024*
Held office hours and graded assignments.

Technical University of Denmark Kongens Lyngby, Denmark
Co-organizer, Special Course on Reinforcement Learning *Jan 2021*
Co-organized a three-week, full-time special course with Ole Winther.

Technical University of Denmark Kongens Lyngby, Denmark
Teaching Assistant, 02456: Deep Learning *Fall 2019, Fall 2020*
Significant course material contributions; supervised class projects of 100+ students through weekly meetings.

Technical University of Denmark Kongens Lyngby, Denmark
Teaching Assistant, 02454: Introduction to Cognitive Science *Fall 2019*
Assisted in tutorial sessions and graded assignments.

Invited Talks

UC San Diego, TILOS AI Institute	"What Makes a Good World Model?"	Oct 2026
Foundation of German Business (SdW)	"What Makes a Good World Model?"	Oct 2026
NYU Global AI Frontier Lab	"What Makes a Good World Model?"	Oct 2026
World Models Workshop, SAB	"What Makes a Good World Model?"	Oct 2026
Mila – Quebec AI Institute	"Hallucination in World Models is Predictable and Preventable"	Oct 2026
Sony AI	"Improving Agents via Online Interaction"	May 2026
University of Michigan	"Robots That Learn by Doing"	Apr 2026
UC Riverside	"AI Agents Should Learn by Doing"	Apr 2026
Carnegie Mellon, Intent Lab	"Improving Agents via Online Interaction"	Feb 2026
University of Washington	"Improving Agents via Online Interaction"	Feb 2026
Johns Hopkins University	"Improving Agents via Online Interaction"	Feb 2026
Stanford, REAL Lab	"Improving Agents via Online Interaction"	Feb 2026
NYU, CILVR Lab	"Improving Agents via Online Interaction"	Feb 2026
UC Berkeley, Robot Learning Lab	"Building Massively Multitask World Models"	Jan 2026
Embodied World Models Workshop, NeurIPS	"Massively Multitask World Models for Continuous Control"	Dec 2025
World Models Workshop, ICCV	"Massively Multitask World Models for Continuous Control"	Oct 2025
Cohere Labs	"World Models on an Academic Budget"	Aug 2025
Chinese University of Hong Kong	"Learning World Models for Embodied AI"	May 2025
World Models Workshop, ICLR	"World Models on a Budget"	Apr 2025
UIUC, CSL Student Conference	"Learning World Models for Robots"	Feb 2025
Copenhagen School of Design and Technology	"Data-Driven Algorithms for Robotics"	Dec 2024
University of Southern California	"Data-Driven World Models for Robots"	Oct 2024
Hugging Face	"TD-MPC, TD-MPC2, and Beyond"	Jun 2024

BeNeRL Reinforcement Learning Seminar	"Data-Driven World Models for Robots"	Jun 2024
UC San Diego, TILOS AI Institute	"Large Datasets and Models for Robots in the Real World"	May 2024
University of Michigan	"Robot Learning with (Generalist) World Models"	Jan 2024
Georgia Tech	"Building Generalist World Models"	Jan 2024
Technical University of Denmark	"Data-Driven World Models at Scale: Why, What, and How?"	Dec 2023
Tsinghua University, IIIS	"The Next Generation of World Models"	Mar 2023
Mila and ServiceNow Research	"World Models with Behavioral Priors"	Feb 2023
Georgia Tech	"Towards Sample-Efficient Robot Learning with World Models"	Jan 2023
Meta AI (FAIR)	"Pretraining for Control: Current Challenges and Solutions"	Jan 2023
Delft University of Technology	"A Path Towards Generalist Agents"	Oct 2022
Generally Intelligent podcast	Episode 25	Dec 2022
Intel AI	"Temporal Difference Learning for Model Predictive Control"	Apr 2022
Intel AI	"Robots That Generalize"	Aug 2021
G-Research	"Agents That Generalize and Adapt"	Feb 2021

Service

Workshop Organizer

Workshop on Bimanual Dexterous Tool Use (CoRL)	2026
Pretrain to Adapt Workshop (CoRL), Program Chair	2026
Embodied World Models Workshop (NeurIPS)	2026
2nd Workshop on World Models (ICLR)	2026

Workshop Area Chair

Foundations of LLM Post-Training in Changing Environments (NeurIPS)	2026

Conference Reviewer

Conference on Neural Information Processing Systems (NeurIPS)	2022, 2023, 2024, 2025, 2026
International Conference on Machine Learning (ICML)	2022, 2023, 2025, 2026
International Conference on Learning Representations (ICLR)	2024, 2025, 2026, 2027
Conference on Computer Vision and Pattern Recognition (CVPR)	2022, 2023, 2026
European Conference on Computer Vision (ECCV)	2022, 2026
International Conference on Computer Vision (ICCV)	2023
International Conference on Robotics and Automation (ICRA)	2023, 2024, 2026
Conference on Robot Learning (CoRL)	2025, 2026
International Conference on Intelligent Robots and Systems (IROS)	2023, 2025, 2026
AAAI Conference on Artificial Intelligence (AAAI)	2021

Journal Reviewer

IEEE Robotics and Automation Letters (RA-L)	2021, 2022, 2023, 2024, 2025
IEEE Transactions on Robotics (T-RO)	2024, 2026
Robotics and Autonomous Systems (RAS)	2026
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	2024
Journal of Machine Learning Research (JMLR)	2023
International Journal of Computer Vision (IJCV)	2023

Workshop Reviewer

Workshop on Modeling Uncertainty in Robotic World Models (CoRL)	2026
Workshop on X-Embodiment Robot Learning (CoRL)	2024
Foundation Models for Decision-Making (NeurIPS)	2023
Self-Supervised Learning: Theory and Practice (NeurIPS)	2022, 2023
Learning Dexterous Manipulation (RSS)	2023
Structural and Compositional Learning on 3D Data (CVPR)	2023
Generalizable Policy Learning in the Physical World (ICLR)	2022

Mentoring Programs and Community

UC San Diego CSE PhD Visit Days, co-organizer	2026
NeurIPS 2025, volunteer	2025
Queer in AI, reviewer for graduate application microgrants	2024
UC San Diego Graduate Women in Computing, mentor for first-year PhD students	2024
UC San Diego GradAMP, mentor for prospective PhD applicants	2022, 2023